

KWOK KEITH

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www.linkedin.com/in/kwokkeith · https://kwokkeith.github.io/Portfolio

Education

Singapore University of Technology and Design (SUTD)

Bachelor of Engineering in Computer Science

- IES Gold Recipient - Top Student in Cohort
- Honors with Highest Distinction
- Director's Honor List 2021/22, 2022/23, 2023/24

Singapore

May 2025

Reykjavik University

Full-time exchange student

Reykjavik, Iceland

Spring 2024

Stanford University

Full-time exchange student

Palo Alto, CA

Summer 2022

Experience

Hewlett Packard Inc.

Automation Software Developer Intern

Singapore

September 2023 - December 2023

- Developed end-to-end software solutions such as Image to Text application using Tesseract Optical Character Recognition as well as Regulatory Web Documentation using Ruby and Sass Framework.
- Developed a live language translation tool powered by AI for meetings with global stakeholders by bench-marking different models such as WhisperAI, Helsinki, Mistral-OpenOrca and seamlessM4T.

Singapore University of Technology and Design

Undergraduate AI Researcher

Singapore

August 2023 - March 2024

- Worked closely together with a team of 5 researchers to understand the impact of Large Language Models (LLMs) and their effectiveness on student's learning.
- Analyzed and generated insights from qualitative data that has shown various student behaviors and habits when using GPT for their learning.

Project

Changi Airport Group - Spot Cleaning Robot

Full-Stack Robotics

SUTD

September 2024 - April 2025

- Developed an AI-driven autonomous spot cleaning robot using ROS Noetic, featuring real-time litter detection and intelligent spot cleaning for Singapore's Changi Airport.
- Implemented full navigation stack (i.e. mapping, localization, path planning, and behavior control)
- Built the supporting software stack including a web UI, backend database integration, and custom litter detection algorithm.

American Sign Language Translation

Machine Learning

SUTD

January 2025 - April 2025

- Designed a deep neural network incorporating MediaPipe, binary classifier for dynamic or static actions, Long-Short Term Memory, Convolutional Neural Network and ending with a Large Language Model for structured English translations from American Sign Language.

Raft Consensus

Software Development

Reykjavik University

April 2024 - May 2024

- Implemented the Raft Consensus algorithm for distributed and concurrent programming in GO.

Finance Automation

Software Development

SUTD

June 2023 - August 2023

- Led the development of a full-stack application that extracts text from receipts using Tesseract OCR.
- Automated submission to SUTD's SAP Concur system using authenticated browser sessions and HTTP requests.
- Built a C#/.NET MAUI database-integrated application with native file system support.

Skills

Technical Languages

Python, C#, Java, Javascript, Haskell, GO, Rust, C, C++

Systems and Frameworks

.NET Core, ROS, Git, Linux, React, Unity, FPGA, Lucid HDL